

Module 4: Network Layer & Routing

IPv4 20-Byte Header, CIDR Subnetting Calculations, IPv6, Distance Vector (Bellman-Ford) & Link State (Dijkstra)

Module IV: Network Layer, IP Addressing & Routing Algorithms

1. Network Layer Service Models: Virtual-Circuit vs. Datagram Packet Switching
2. IPv4 Header Format: 20-Byte Field Breakdown & Fragmentation Fields (Offset, Flags)
3. Classful IP Addressing (Classes A, B, C, D, E & Special Loopback / Private Blocks)
4. Subnetting & Classless Inter-Domain Routing (CIDR) Slash Notation ($/n$)
5. Longest Prefix Match Algorithm in Forwarding Tables with Worked Lookups
6. IPv6 Protocol Architecture: 40-Byte Fixed Base Header & Extension Headers
7. IPv4 to IPv6 Transition Mechanisms: Dual-Stack, Tunneling & Header Translation
8. Unicast Routing Principles: Intra-AS (IGP) vs. Inter-AS (EGP) Autonomous Systems
9. Distance Vector Routing (DVR): Bellman-Ford Equation & Count-to-Infinity Problem
10. Solutions to Count-to-Infinity: Split Horizon & Poisoned Reverse Mechanisms
11. Link State Routing (LSR): Link State Advertisements & Dijkstra's Algorithm Tree
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 2 – 4: IPv4 Datagram Format & CIDR Subnetting

IPv4 Header Field	Bit Width	Functional Purpose & Transmission Invariant
Version	4 bits	Identifies IP version (`0100` for IPv4).
Header Length (IHL)	4 bits	Length of header in 32-bit (4-byte) words. Minimum value is 5 (20 bytes).
Total Length	16 bits	Entire datagram length (Header + Data) in bytes. Max size = 65,535 bytes.
Identification, Flags, Offset	16 + 3 + 13 bits	Handles MTU packet fragmentation. Fragment Offset is measured in 8-byte units. `DF` = Don't Fragment, `MF` = More Fragments.
Time to Live (TTL)	8 bits	Hop counter decremented by 1 at each router to prevent infinite forwarding loops. Discarded with ICMP Time Exceeded when $TTL = 0$.
Protocol	8 bits	Demultiplexing to transport layer (`6` for TCP, `17` for UDP, `1` for ICMP).
Header Checksum	16 bits	1's complement sum of header fields recalculated at every router as TTL decrements.

Step-by-Step Solved Problem: CIDR Subnet Allocation

Problem: An organization is assigned the network block `200.10.20.0/24`. It needs to divide this block into 4 equal subnets for 4 departments.

1. **Step 1: Determine Additional Subnet Bits (k):**

$$2^k \geq 4 \implies k = 2 \text{ bits}$$

New Subnet Mask: $/24 + 2 = /26 \implies \mathbf{255.255.255.192}$.

2. **Step 2: Compute Total and Usable Host Addresses per Subnet:**

$$\text{Total Addresses per Subnet} = 2^{32-26} = 2^6 = \mathbf{64}$$

$$\text{Usable Host Addresses} = 64 - 2 = \mathbf{62} \quad (\text{subtracting Network ID and Directed Broadcast})$$

3. **Step 3: Subnet Allocation Table:**

Subnet	Network Address	Usable Host IP Range	Broadcast Address
Dept 1	`200.10.20.0/26`	`200.10.20.1` - `200.10.20.62`	`200.10.20.63`
Dept 2	`200.10.20.64/26`	`200.10.20.65` - `200.10.20.126`	`200.10.20.127`
Dept 3	`200.10.20.128/26`	`200.10.20.129` - `200.10.20.190`	`200.10.20.191`
Dept 4	`200.10.20.192/26`	`200.10.20.193` - `200.10.20.254`	`200.10.20.255`

Topic 9 – 11: Routing Algorithms (Distance Vector vs. Link State)

Feature	Distance Vector Routing (DVR — RIP)	Link State Routing (LSR — OSPF)
Mathematical Algorithm	Bellman-Ford Equation: $D_x(y) = \min_v \{c(x, v) + D_v(y)\}$	Dijkstra's Shortest Path Algorithm: Computes full shortest-path tree from source.
Information Exchanged	Vector of estimated distances to all destinations.	Exact state and cost of directly attached links (LSA).
Destination of Updates	Sent only to directly adjacent neighbors.	Flooded globally to all routers in the entire network area.
Convergence & Stability	Slow convergence; prone to Count-to-Infinity routing loops.	Fast convergence; completely immune to routing loops.

Q1. Explain the Count-to-Infinity problem in Distance Vector Routing and how Split Horizon with Poison Reverse resolves it. (8 Marks)

- **Count-to-Infinity Problem:** When a link $A - B$ goes down, router A sets its cost to ∞ . However, neighbor C may still advertise that it has a path to A via B with cost 2. Router B receives this and updates its cost to A as $2 + 1 = 3$, advertising it back to C . C updates to $3 + 1 = 4$, and both routers slowly increment their distance vectors step-by-step up to ∞ (16 in RIP).
- **Split Horizon with Poison Reverse:** If router C routes traffic to destination X through neighbor B , C advertises its distance to X back to B as ∞ (Poisoned Reverse). This prevents B from wrongly routing traffic back through C during link failures.